Beyond the Classroom

My Canadian Tech Journey

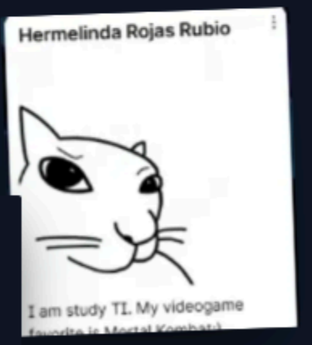
I'm Hermelinda Rojas Rubio, an Information Technology student from Querétaro, Mexico. For four weeks I studied at Cambria
Ontario. This portfo
work I made.

Host college
Cambrian College

Location
Greater Sudbury, Ontario, Canada

I am Hermelinda Rojas Rubio, an Information Technology student from Querétaro, Mexico. For four weeks I studied at Cambrian College in Greater Sudbury, Ontario.

Four weeks · October–November 2026



My first post on the class welcome board

Four weeks in Sudbury, day by day

The program schedule, with the work I made along the way. Teal marks my own evidence; everything else is context from the program.

● My work ● Class material only ○ Program event ◌ Upcoming

Today

Week 1

Introduction · Oct 12 – Oct 18

Read my reflection on week 1

Oct 14 ○ Portfolio Exchange introduced

Oct 14 ● Arduino circuits in Tinkercad

Week 2

Developing · Oct 19 – Oct 25

Read my reflection on week 2

Oct 19 ● Game Design

Oct 19 ● Micro:bits: Cryptography

Oct 20 ● Animation: Blender

Oct 21 ● Harmony rigging · Character posing

Oct 22 ● Requirements and prototyping

Oct 22 ● Game Design

Week 3

Review · Oct 26 – Nov 1

Read my reflection on week 3

Oct 26 ● AI Agents

Oct 27 ● Blender · Harmony rigging

Oct 28 ○ Entrepreneurship workshop

Oct 28 ● Our business idea won

Oct 29 ● Life drawing: gesture and drapery

Oct 29 ● My gesture sketches

Week 4

Presentation · Nov 2 – Nov 7

Read my reflection on week 4

Nov 2 ● Game Design

Nov 2 ● StarFall 2.0: waves, lives, Game Over

Nov 2 ● Database Bootcamp

Nov 2 ● My first SQL queries

Nov 3 ○ Blender · IoT APIs

Nov 3 ● A 3D model in Blender

The exchange followed a four-week schedule. Teal marks the workshops where I have my own evidence. Week four is still ahead.

ony · Sacred Fire and enous teaching

rawing · Game Design

Nov 1 ○ Sudbury Wolves hockey game

Nov 7 ◌ Flight home to Mexico City

Week 1 · Introduction · Oct 12 – Oct 18

A New Beginning: Everything Felt Like a Movie

My first week in Canada. I was excited, nervous and happy, because everything around me was new.

Workshops this week: Networking, Animation, Arduino and Micro:bits

Read the full reflection

"Sometimes I felt like I was in a movie. Every morning, when the bus picked us up and took us to Cambrian College, I would look around and think about how incredible it was that I was actually in Canada."

In my words

Week 2 · Developing · Oct 19 – Oct 25

Creativity Technology and New Expe

I started to feel more comfortable, and I couldn't wait to discover

y good at

everything to learn someth ing

Workshops this week: Game Design, Arduino and Micro:bits, Animation, Drawing

In my words

My first week felt like a movie. Every morning the bus took us to Cambrian College, and I could not believe I was really in Canada.

Week 2 · Developing · Oct 19 – Oct 25

Creativity, Technology and New Experiences

I started to feel more comfortable, and I couldn't wait to discover what we would create in each class.

Workshops this week: Game Design, Arduino and Micro:bits, Animation, Drawing

Read the full reflection

"I don't have to be naturally good at everything to learn something valuable from it."

In my words

Week 3 · Review · Oct 26 – Nov 1

An Idea That Won

In week three, my t
presented it, and ou
pumpkin with lights

ea,
won."

In week two I felt more comfortable. I made a game with stars, loved our micro:bit class, and learned that I don't have to be naturally good at everything to learn something valuable from it.

In my words

Read the

Ten workshops

Each workshop gets its own case study. The numbers count the pieces of evidence I have so far.

By subject | By week

Game Design Case study

Weeks 1, 2, 4

My work (5) Class material (2)

Game design theory in week 1. StarFall, my first game in GameMaker, started in week 2 and grew into version 2.0 in week 4 (waves, three lives, Game Over), with its code and my notes on player psychology

Drawing and Character Development Case study

Weeks 2, 3, 4

My work (4) Class material (2)

A layered digital pose drawing and five thumbnail sketches

2D Animation in Harmony Case study

Weeks 1, 2, 3, 4

My work (2) Class material (1)

Keyframe practice and a walk cycle on a rigged character

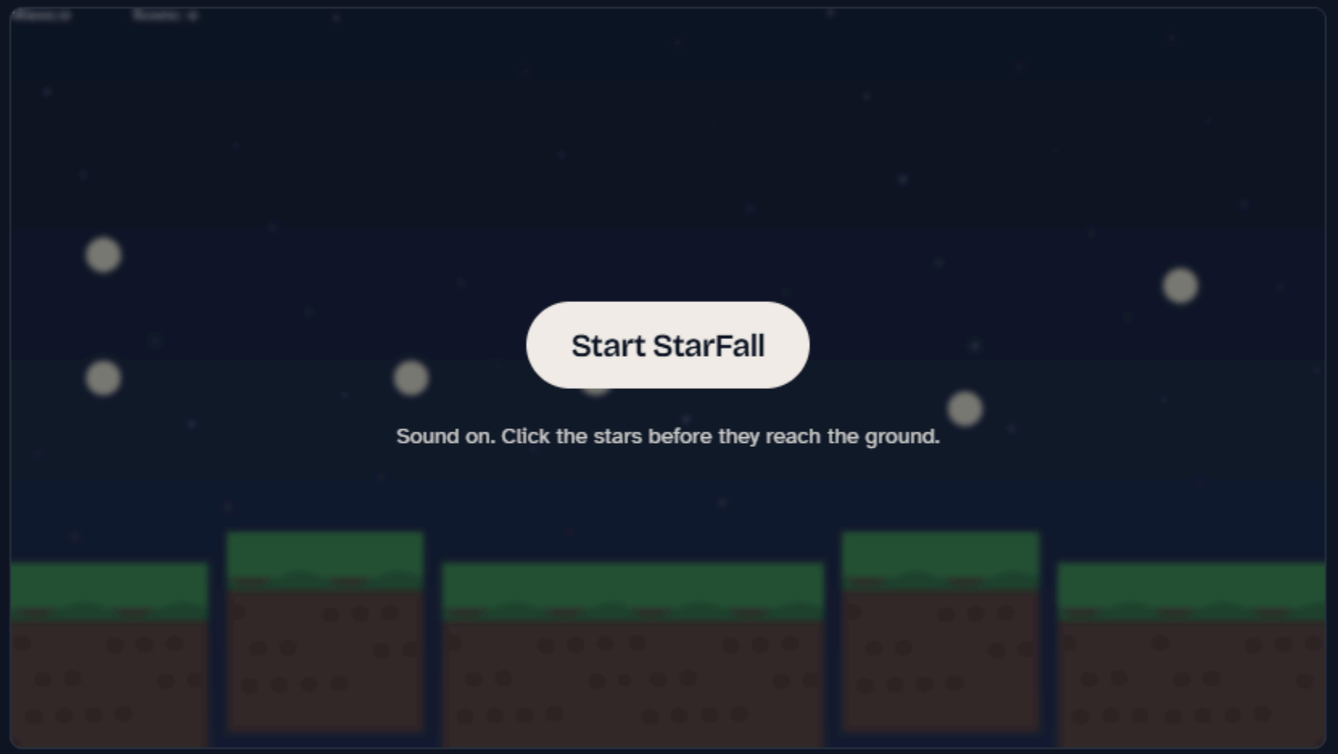
3D Animation in Blender Case study

Weeks 2, 3, 4

My work (1)

My short

There are ten workshops. Each case study is built from my notes, files and photos, and shows clearly what came from the class.



Start | Restart | Music | Sound effects | Volume | Collision boxes

Full screen

StarFall is my first video game, made in GameMaker during the workshop. In this online version, the workshop's sprites and sounds are replaced by placeholders.

How it works

◇ Interactive demonstration built for this portfolio

Version 2.0 of the game is eleven short events. Choose one to see the code from my project and try what it does.

obj_gameManager

- Create — Changed in 2.0
- Step — New in 2.0
- Alarm 0 — Off in 2.0
- Key Press R — New in 2.0
- Draw GUI — Changed in 2.0

obj_star

- Create — Object behaviour
- Step — Game loop
- Left Pressed — Changed in 2.0
- Collision with obj

obj_explosion

- Anima

obj_gameOver

Spawning · New in 2.0

obj_gameManager · Step

Every step it checks two things. When no star is left, it sends a wave with as many stars as the wave number, and the next wave is one bigger. When three stars have been lost, it shows Game Over, plays its sound and stops the music.

Stars falling now



Stars lost

`instance_number(obj_star) = 1 · wave = 2 · starsLost = 0 · gameOver = false · gameScore = 0`

Room start: no stars yet, wave = 1. No star left: wave 1 sends 1 star.

Click a star | Let a star land | Restart (R or click the sign)

In version 2.0 the stars come in waves, each one bigger than the last. Three stars on the ground mean Game Over, and you can start again.

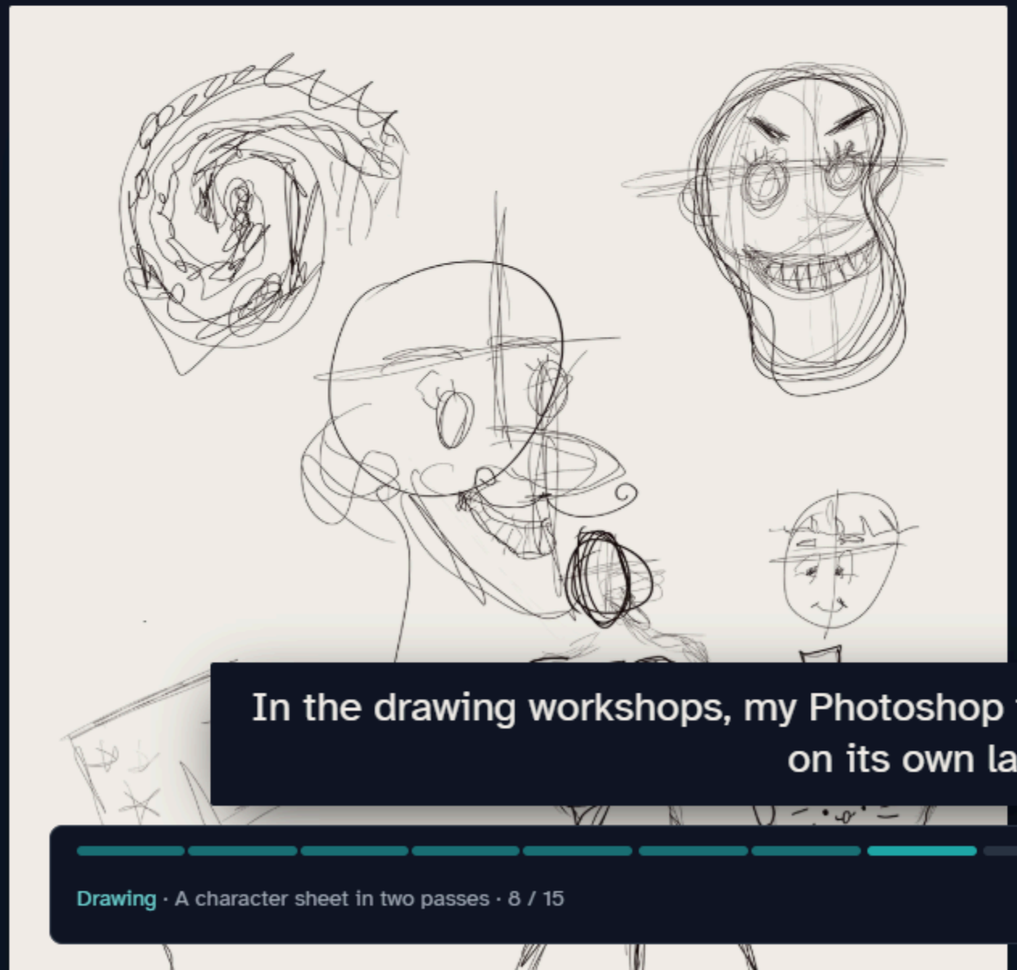
```
frameCount++
```

```
instance_create_depth(random_range(0+50, room_width-50), -150, 1, obj_star)
```

A character sheet in two passes

◇ Interactive demonstration built for this portfolio

My Photoshop file keeps each round of sketching on its own layer. Turn them on and off to see how the page was built.



Layers in the Photoshop file



Layer 2
Second pass



Layer 1
First pass

Layer 0 the paper

Second pass opacity 100%

Point at a layer to isolate it, or use the eye buttons. Each pass adds new characters on its own layer, so earlier ideas stay untouched.

In the drawing workshops, my Photoshop file kept each round of sketching on its own layer.

Keyframes and timing

Recreate the exercise from my photo: drag the circle to set keys, then play it. Compare how Harmony fills the frames in between.

◇ Interactive demonstration built for this portfolio



In the Harmony workshop I worked with keyframes. Stop-motion keys hold a pose; motion keys let the software draw the frames in between.

Distance in, light out

◇ Interactive demonstration built for this portfolio

```
set dis to read ultrasonic distance sensor on trigger pin 6
if dis < 200 then
    set pin 11 to HIGH
if dis < 100 then
    set pin 11 to HIGH
```

The highlighted blocks are the ones that run for the current distance.

● Class demonstration

On the projector, the class built a circuit in Tinkercad: an ultrasonic sensor controls the LEDs. The blocks below match my photo.

pin 13 pin 12 pin 11

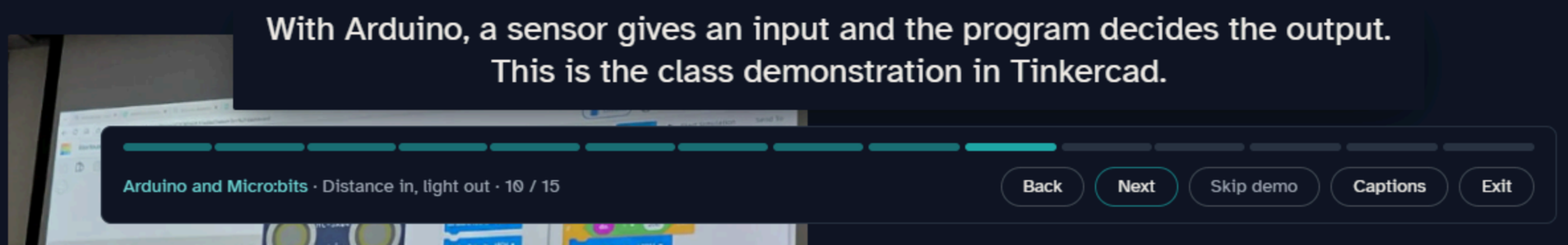
dis
250

Move an object in front of the sensor

Nothing is close: no condition is true, so no block changes pin 11.

Add an else block

My exploration, not part of the class code: an else that sets pin 11 LOW makes the LED turn off again.



Private, unchanged, available

◇ Interactive demonstration built for this portfolio

The class told one story: a nurse sends an urgent instruction. It can be read, changed or blocked. Each harm breaks one part of the CIA triad.



Confidentiality — Who can read it?

Integrity

Availability

The harm: someone changes one instruction

An invoice for $1,200 is altered to $12,000 before delivery.

Sent

```
Invoice total: $1,200
SHA-256 2688216d59319842916bfe08ee639618…
```

Received

```
Invoice total: $1,200
SHA-256 2688216d59319842916bfe08ee639618…
```

Tamper in transit — Fingerprints match: the message is unchanged.

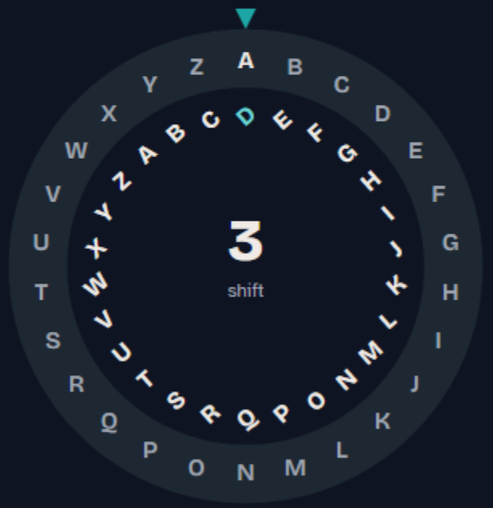
Every security harm breaks one part of the CIA triad. A changed invoice is detected because its fingerprint changes.

Ciphers from my notebook

◇ Interactive demonstration built for this portfolio

Encryption turns plaintext into ciphertext. These are the three ciphers I wrote down: one shift, a shuffled alphabet, and a moving shift.

Caesar | Shuffled alphabet | Vigenère



Plaintext

```
BY HERMELINDA
```

Ciphertext

```
EB KHUPHOLQGD
```

Class exercise: encrypt “BY” and your name with a shift of 3. Shifting wraps around with clock arithmetic: (letter + shift) mod 26.

Shift 3

These ciphers come from my notebook. With the key LEMON, the word ATTACK becomes L X F O P V.

The approved request flow

◇ Interactive demonstration built for this portfolio

Release 1's process, from Draft to a decision. Play each role and watch which requirements every step satisfies.

Draft | Pending Approval | Returned for Correction | Approved | Denied | Cancelled

Employee

- [ ] Fields complete and dates valid

Submit

Correct and resubmit

Cancel request

Supervisor

Approve

Deny

Return

HR

- [ ] Supervisor missing

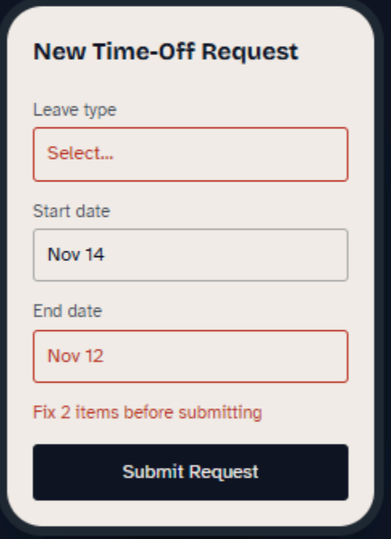
Route manually

Start a new request

A new request starts in Draft.

REQ-005

Path so far: Draft



Low-fidelity screen, after the class design example

In business analysis the class worked on Northstar, a fictional case. Every step of the approval flow traces back to a requirement.

The agent loop

◇ Interactive demonstration built for this portfolio

Every agent runs the same heartbeat: think, act, observe, and repeat until the goal is met. Step through the example from class.



Goal	Do I need a jacket in Sudbury today?
Think	I need today's weather → use the weather tool
Act	`get_weather("Sudbury")`
Observe	4 °C, light snow
Think	Cold and snowy → I can answer now
Done	Yes! 4 °C with light snow. Bring a warm jacket.

Back | Next step

The AI Agents workshop explained how an agent thinks, acts and observes until its goal is met. This section is class material.

Anato…g of an agent

…ch part around the brain was the topic of one lecture.

◇ Interacti…

Greater Sudbury, Ontario

Cambrian College is in Greater Sudbury, in Northern Ontario. It is about 3,370 km from Querétaro in a straight line. For four weeks, it was my campus.



Greater Sudbury is about 3,370 kilometres from Querétaro. The Database and IoT workshops are still ahead in week four. Thank you.